

MINI PROJECT PORTFOLIO

2D GRAPHICS EDITOR USING C

Submitted By

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ABSTRACT

The 2D Graphics Editor is a menu-driven application developed using the C programming language. The project provides a text-based graphical environment where users can create and manipulate basic geometric shapes such as rectangles, lines, triangles, and circles on a 20×40 canvas. The editor supports object creation, deletion, modification, display, and canvas reset operations.

OBJECTIVES

1. Create a 2D drawing canvas using a character array.
2. Implement drawing of geometric shapes.
3. Support deletion and modification of objects.
4. Display graphical output in the console.
5. Apply modular programming concepts in C.

PROBLEM STATEMENT

Design and implement a menu-driven 2D Graphics Editor using the C programming language.

SYSTEM REQUIREMENTS

Hardware: Intel Core i3+, 4 GB RAM.

Software: Windows 10/11, C, Code::Blocks, GitHub.

SYSTEM DESIGN

Canvas Size: 20 rows × 40 columns. Coordinate range x:0-39, y:0-19. Top-left corner is (0,0).

DATA STRUCTURES USED

Object Types: Rectangle, Line, Triangle, Circle.

Canvas: 20×40 character array.

Object Storage: Array of objects.

FEATURES IMPLEMENTED

Canvas Initialization, Display Canvas, Draw Rectangle, Draw Line, Draw Triangle, Draw Circle, Delete Object, Modify Object, Reset Canvas, Exit Application.

ALGORITHM

Initialize canvas, display menu, accept user choice, perform operation, render objects, display canvas, repeat until exit.

IMPLEMENTATION DETAILS

Rectangle, Line, Triangle, and Circle drawing implemented using coordinates and dimensions. Circle uses $(x-cx)^2 + (y-cy)^2 \leq r^2$.

TEST CASES

Rectangle(5,5,10,5), Line(2,2,8), Triangle(20,10,6), Circle(30,10,4), Delete Object, Modify Object.

OUTPUT SCREENSHOTS

Insert screenshots of menu, rectangle, line, triangle, circle, delete and reset operations.

```
--- Main Menu ---
1. Draw Rectangle
2. Draw Line
3. Draw Triangle
4. Draw Circle
5. Delete Object
6. Modify Object (Move)
7. Display Canvas
8. Reset Canvas
9. Exit
Select an option (1-9):
```

```
Horizontal Line added successfully!
0123456789012345678901234567890123456789
0-----
1-----
2_*****_
3-----
4-----
5-----
6-----
7-----
8-----
9-----
10-----
11-----
12-----
13-----
14-----
15-----
16-----
17-----
18-----
19-----
```

```

Right-angled Triangle added successfully!
0123456789012345678901234567890123456789
0-----
1-----
2-----
3-----
4-----
5-----
6-----
7-----
8-----
9-----
10-----
11-----
12-----
13-----
14-----
15-----
16-----
17-----
18-----
19-----

```

```
Circle added successfully!
```

```
0123456789012345678901234567890123456789
0-----
1-----
2-----
3-----
4-----
5-----
6-----
7-----
8-----
9-----
10-----
11-----
12-----
13-----
14-----
15-----
16-----
17-----
18-----
19-----
```

```

Success: Deleted Object ID 5 (Type 4) found at (30, 10).
0123456789012345678901234567890123456789
0 -----
1 -----
2 -----
3 *****
4 -----
5 -----
6 -----
7 -----
8 -----
9 -----
10 -----
11 -----
12 -----
13 -----
14 -----
15 -----
16 -----
17 -----
18 -----
19 -----

```

```

Canvas and all objects have been reset.
0123456789012345678901234567890123456789
0 -----
1 -----
2 -----
3 -----
4 -----
5 -----
6 -----
7 -----
8 -----
9 -----
10 -----
11 -----
12 -----
13 -----
14 -----
15 -----
16 -----
17 -----
18 -----
19 -----

```

ADVANTAGES

Simple interface, demonstrates C concepts, modular design, supports multiple objects.

FUTURE ENHANCEMENTS

Colored objects, filled shapes, save/load drawings, GUI implementation.

CONCLUSION

The project successfully implements a 2D Graphics Editor using C with object creation, modification, deletion, and display capabilities.

REFERENCES

Kernighan & Ritchie - The C Programming Language;

Code::Blocks Documentation;

GCC Documentation;

GitHub Documentation.